

4/4/19

* Asymptotic performance of estimators —

(11)

Motivation: what happens to the estimate as the # sample increases —

Consistency:

Defn let $w_n: \mathcal{X}^n \rightarrow \Theta$ be an estimator for any $n \geq 1$

($(w_n)_{n \geq 1}$ is a sequence of estimators)

$(w_n)_{n \geq 1}$ is said to be consistent if —

$$\lim_{n \rightarrow \infty} P_{X_1, \dots, X_n \sim P_\theta} [|w_n(X_1, \dots, X_n) - \theta| > \epsilon] = 0 \quad \forall \epsilon$$

In other words, under $X_1, \dots, X_n \sim P_\theta, \forall \theta \in \Theta$

$$w(X_1, \dots, X_n) \xrightarrow{P} \theta$$

eg $X_1, \dots, X_n \sim N(0, 1)$

$$w_n(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{is consistent} \quad \left(\begin{array}{l} \text{weak law} \\ \text{of large} \\ \text{nos} \end{array} \right)$$

$$(\text{Var} [w_n(X_1, \dots, X_n)] = \gamma_n)$$

Remark : if $w_n(X_1, \dots, X_n) \xrightarrow{\text{mean square}} \theta, \forall \theta \in \Theta$
then $(w_n)_{n \geq 1}$ is consistent.

$$E_\theta [(w_n(X_1, \dots, X_n) - \theta)^2] = \text{Var}_\theta w_n + (\text{Bias}_\theta(w_n))^2$$

Hence if the bias and

Variance of w_n asymptotically vanish, then

$(w_n)_{n \geq 0}$ is Consistent.

→ Consistency of the ML estimator -

Recall that MLE : Given $X_1, \dots, X_n \sim P_\theta, \theta \in \Theta$

$$\hat{\theta}_n^{\text{MLE}} = \arg \max_{\theta \in \Theta} \sum_{i=1}^n \log P_\theta(x_i)$$

eg : $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$ or $\stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$

$$\hat{\theta}_n^{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \text{so } (\hat{\theta}_n^{\text{MLE}})_{n \geq 1} \text{ is } \underline{\text{Consistent}} -$$

In General, the MLE to be Consistent?
(Yes!)

Intuition : Suppose $\theta_0 \in \Theta \subseteq \mathbb{R}$

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_{\theta_0}, \quad \theta_0 \in \mathbb{R}$$

$\Theta =$

$$\hat{\theta}_n^{\text{MLE}} = \arg \max_{\theta \in \mathbb{R}} \underbrace{\frac{1}{n} \sum_{i=1}^n \log P_\theta(x_i)}_{\substack{L(x, \theta) \\ | \\ (x_1, \dots, x_n)}}$$

If $L(x, \theta)$ is differentiable w.r.t θ

$$\Rightarrow \frac{\partial}{\partial \theta} L(x, \theta) \Big|_{\theta = \hat{\theta}_n^{\text{MLE}}} = 0$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n \left. \frac{\partial}{\partial \theta} \log P_{\theta}(x_i) \right|_{\theta = \hat{\theta}_n^{MLE}(x_1, \dots, x_n)} = 0 \quad - (i) \quad (72)$$

for any $\theta' \in \Theta$, $x_1, x_n \sim P_{\theta_0}$

$$\frac{1}{n} \sum_{i=1}^n \left. \frac{\partial}{\partial \theta} \log P_{\theta}(x_i) \right|_{\theta = \theta'}$$

$n \rightarrow \infty$

$$\underbrace{\left\{ \left. \left[\frac{\partial}{\partial \theta} \log P_{\theta}(x_1) \right] \right|_{\theta = \theta'} \right\}}_{=: J(\theta_0, \theta')}$$

In particular, for $\theta' = \theta_0$,

$$J(\theta_0, \theta_0) = \left\{ \left. \left[\frac{\partial}{\partial \theta} \log P_{\theta}(x_1) \right] \right|_{\theta = \theta_0} \right\}$$

$$= \int_{\mathcal{X}} \left. \left[\frac{1}{P_{\theta_0}(x)} \frac{\partial}{\partial \theta} P_{\theta}(x) \right] \right|_{\theta = \theta_0} P_{\theta_0}(x) dx$$

$$= \left(\left. \left[\frac{\partial}{\partial \theta} \int_{\mathcal{X}} P_{\theta}(x) dx \right] \right|_{\theta = \theta_0} \right) \quad \left| \begin{array}{l} \text{interchanging} \\ \int \text{ and } \partial/\partial \theta \end{array} \right.$$

$$= 0$$

Therefore

$$\theta' = \theta_0 \text{ is a root of } J(\theta_0, \theta') = 0$$

Suppose that θ_0 is the unique root $J(\theta_0, \theta') = 0$
and $J(\theta_0, \theta')$, $l(x, \theta')$ are smooth function of θ' .

$$\nabla \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log P_{\theta}(x_i) \Big|_{\theta = \hat{\theta}_n^{MLE}} = 0$$

$$\nabla \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log P_{\theta}(x_i) \Big|_{\theta = \theta_0} \simeq J(\theta_0, \theta_0) = 0$$

$$(\theta_0, \hat{\theta}_n^{MLE} \simeq \theta_0 \text{ for } n \text{ large})$$

approximate
become better
better when n
b/c very large

Theorem: (Consistency of MLE, Poor - prob IV-D.1)

Suppose $X_1, \dots, X_n \stackrel{iid}{\sim} P_{\theta}$, $\theta \in \Theta \subseteq \mathbb{R}$

and $\hat{\theta}_n^{MLE}$ is an MLE. Moreover, Suppose:

(1) $\forall x: \log P_{\theta}(x)$ is differentiable in θ

$$J(\theta_1, \theta_2) = E_{\theta_1} \left[\frac{\partial}{\partial \theta} \log P_{\theta}(x) \Big|_{\theta = \theta_2} \right]$$

exists and is finite -

(2) $J(\theta_1, \theta_2)$ is a continuous function of θ_2 , has a unique root at $\theta_2 = \theta_1$, at which point it changes sign.

(3) $\frac{\partial}{\partial \theta} \log P_{\theta}(x) \Big|_{\theta = \theta_2}$ - function of θ_2

(4) $\forall n \geq 1$, $\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log P_{\theta}(x_i) \Big|_{\theta = \theta_2}$ has a unique root $\hat{\theta}_n^{MLE}$

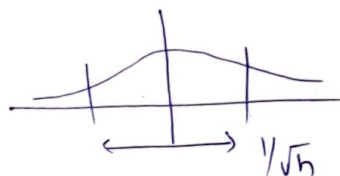
Then $\hat{\theta}_n^{MLE} \xrightarrow{P} \theta_0$ as $n \rightarrow \infty$

* Efficiency of the MLE ÷ (Asymptotic Normality)
 So far we have seen $\hat{\theta}_n^{MLE} = \theta_0$, as $n \rightarrow \infty$ (13)

What about $|\hat{\theta}_n^{MLE} - \theta_0|$?
 as a function of n

eg1) $X_1, \dots, X_n \stackrel{iid}{\sim} N(\theta_0, 1)$

$$\begin{aligned} & \frac{1}{n} \sum_{i=1}^n X_i - \theta_0 \\ &= \frac{1}{n} \left(\sum_{i=1}^n (X_i - \theta_0) \right) \sim N(0, 1/n) \end{aligned}$$



eg2) $X_1, \dots, X_n \sim \text{Ber}(\theta)$

$$\frac{1}{n} \sum_{i=1}^n (X_i - \theta_0) \longrightarrow$$

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n (X_i - \theta_0) \xrightarrow[\text{Limit theorem } (n \rightarrow \infty)]{\text{Central}} N(0, \text{Var}(X_1))$$

$$\frac{N(0, \text{Var}(X_1))}{\sqrt{n}}$$

→ let $X_1, \dots, X_n \stackrel{iid}{\sim} P_{\theta_0}$, $\theta_0 \in \Theta \subseteq \mathbb{R}$. $\hat{\theta}_n^{MLE}$ denote the MLE of θ based on $X_1, \dots, X_n$. Under suitable conditions on $\{P_\theta : \theta \in \Theta\}$ we have,

$$\sqrt{n}(\hat{\theta}_n^{MLE} - \theta_0) \xrightarrow{d} N(0, v(\theta_0))$$

where $v(\theta_0) = \frac{1}{E_{\theta_0} \left[\left(\frac{\partial}{\partial \theta} \log P_{\theta}(x_1) \right)^2 \right]}$

reciprocal of

is the fisher info of a $\underset{\text{single}}{n}$ sample at θ_0 .

$\Rightarrow$ for large n ,

$$\begin{aligned} E_{\theta_0} \left[(\hat{\theta}_n - \theta_0)^2 \right] &\approx \frac{v(\theta_0)}{n} \\ &= \frac{1}{n F I_{\theta}(x_1)} \end{aligned}$$

which is the
Cramer Rao lower
bound for unbiased
estimation of θ_0 .

idea behind the proof:

Recall: $\hat{\theta}_n = \arg \max_{\theta} L(x, \theta)$

$$L'(x, \hat{\theta}_n^{MLE}) = 0$$

$$\left. \frac{\partial}{\partial \theta} L(x, \theta) \right|_{\theta = \hat{\theta}_n^{MLE}}$$

for any $\theta' \in \Theta$, for any $X \equiv x_1, \dots, x_n$

$$L'(x, \theta') = L'(x, \theta_0) + \frac{(\theta' - \theta_0) L''(x, \theta_0)}{1!} + \frac{(\theta' - \theta_0)^2 L'''(x, \theta_0)}{2!} + \dots$$

at $\theta' = \hat{\theta}_n^{MLE}$

$$LHS = 0 \approx RHS = L'(x, \theta_0) + (\hat{\theta}_n^{MLE} - \theta_0) L''(x, \theta_0)$$

$$\sqrt{n}(\hat{\theta}_n^{MLE} - \theta_0) \approx \frac{-L'(x, \theta_0)}{L''(x, \theta_0)} \times \frac{1}{\frac{1}{\sqrt{n}}}$$

as $n \rightarrow \infty$

(14)

$$(i) \quad \frac{1}{n} l''(x, \theta_0) = \frac{1}{n} \sum_{i=1}^n \left. \frac{\partial^2 \log P_{\theta}(x_i)}{\partial \theta^2} \right|_{\theta = \theta_0}$$

$$\begin{aligned} & \xrightarrow[n \rightarrow \infty]{\text{a.s.}} E_{\theta_0} \left[\frac{\partial^2 \log P_{\theta}(x_i)}{\partial \theta^2} \right] \\ & \quad \left(\begin{array}{l} \text{Strong law} \\ \text{of large no.} \end{array} \right) \\ & = - E_{\theta_0} \left[- \frac{\partial^2 \log P_{\theta}(x_i)}{\partial \theta^2} \right] \\ & = - E_{\theta_0} \left[\left(\frac{\partial \log P_{\theta}(x_i)}{\partial \theta} \right)^2 \right] \\ & = - \frac{1}{V(\theta_0)} \end{aligned}$$

$$(ii) \quad \frac{1}{\sqrt{n}} l'(x, \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left. \frac{\partial \log P_{\theta}(x_i)}{\partial \theta} \right|_{\theta = \theta_0} \quad \begin{array}{l} \xrightarrow{\quad} E_{\theta_0} \left[\frac{\partial \log P_{\theta}(x_i)}{\partial \theta} \right]_{\theta = \theta_0} \\ = 0 \end{array}$$

$$\begin{aligned} & = \xrightarrow[\text{CLT}]{d} - N \left(0, E_{\theta_0} \left(\frac{\partial \log P_{\theta}(x_i)}{\partial \theta} \right)^2 \right) \\ & = - N \left(0, \frac{1}{V(\theta_0)} \right) \end{aligned}$$

Hence

$$= \underline{\underline{N(0, V(\theta_0))}}$$